

Name:		 MANIPAL UNIVERSITY JAIPUR	
Enrolment No:			
Odd Semester End Term Examination, November 2024 Faculty of Engineering, School of Information, Security and Data Science Department of Information Technology B.Tech. – Information Technology SET-B			
Course Code: INT2121		Course: Python Programming	Semester: III
Time: 03 hrs.			Max. Marks: 80
Instructions: All questions are compulsory. Missing data, if any, may be assumed suitably. Calculator is allowed.			
SECTION A (Memory Based Questions)			
S. No.		Marks	CO
Q 1	Write a python expression to calculate the value of the following formula: $x^3 + 2xy - y^2$ use the variables $x = 5$ and $y = 3$. Print the result.	[2]	INT2121.1
Q 2	Write a python code that takes an integer $x = 32$ and a string $y = "3.14"$. Convert the followings: a) x to a string and concatenate it with y . b) y to a float and add it to x	[2]	INT2121.1
Q 3	What is the use of the split() method in python strings?	[2]	INT2121.1
Q 4	Consider the following code: <pre>class Car: def __init__(brand): b = brand my_car = Car("Tesla") print(my_car.b)</pre> What will be the output of the code: a) Tesla b) Car c) brand d) Error	[2]	INT2121.4
Q 5	Given the tuple, $data = (3, 7, 2, 8, 4)$, write a python expression to find the maximum value in the tuple.	[2]	INT2121.2
SECTION B (Concept Based Questions)			
Q 6	a) Differentiate the following with suitable example in python: 1. r+ mode and w+ mode 2. wb+ mode and ab+ mode b) Write a python program to read a file line by line and store it into a list. Find the line number whose length is maximum.	[3+5]	INT2121.3

Q 7	a) Display the following pattern: <pre> 1 2 4 6 3 5 7 9 11 8 10 12 14 16 18 20 </pre> a) Differentiate for loop from while loop in Python Programming with suitable examples?	[5+3]	INT2121.1
Q 8	Consider a class named 'Football' with below given member variables (properties) and method (behavior): (Player_id: integer, player_name: String, match_goals: List of integers (for 5 matches), qualified: Boolean) Define a class method as 'findQualification()' which finds the qualification for the player based on his goals and return boolean <i>status</i> value as mention below: (i) status is <i>false</i> (DISQUALIFIED) if average goals are below 3. (ii) status is <i>true</i> (QUALIFIED), otherwise. Write python code to create the class named 'Football' as per the above description. a) Create two <i>Football</i> type instances (objects) for players of your choice and assign values to all the properties using <i>_init()</i> method. b) Demonstrate how you can find status of each player (qualified or disqualified) based on instances created in Part (a) above.	[8]	INT2121.4
Q 9	a) Differentiate different exceptional handling blocks with suitable examples. b) Write a python program to handle a <i>ZeroDivisionError</i> exception when dividing a number by zero.	[3+3]	INT2121.3
SECTION-C (Analytical Based Questions)			
Q 10	a) Write a python program to calculate the frequency of each element in a list and sort the list by frequency. Input: [4, 5, 6, 5, 4, 3, 5, 2, 4, 6, 3, 3, 2, 1] Expected Output: [1, 6, 6, 2, 2, 4, 4, 4, 5, 5, 5, 3, 3, 3] b) Write a python program to find the sum of a below given matrix's elements without using built-in function. <pre> 2 3 4 5 3 1 1 2 1 </pre>	[5+5]	INT2121.2
Q 11	You are given a 2D NumPy array representing the sales data of a store. Each row represents the sale of each day, and the column represents the product. Each row corresponds to a specific day, and each column represents the sales of a specific product. a) Write a Python program to calculate the total sales for each product across all	[10]	INT2121.5

	days. b) Identify the day with the highest total sales. c) Determine the product with the highest average daily sales. d) Calculate the sales percentage for each day. Input Sales Data: [[200, 150, 300], [400, 100, 250], [300, 350, 200],]		
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SECTION-D (Case Study/Application based Question)

Q 12	Consider the below given <i>Car</i> schema/structure of a dictionary: Movie {C-Name: string, Average: integer, C-Type: string, C-price: integer, Profit: float} a) Write a python code to create <i>Car</i> dictionary to maintain given cars information (take user input). <table><tr><th>C-Name</th><th>Average (in KM/Litre)</th><th>C-Type</th><th>C-price (in lakhs)</th><th>Efficiency (per month)</th></tr><tr><td>Shift Desire</td><td>42</td><td>Deluxe</td><td>13</td><td>0.0</td></tr><tr><td>Audi</td><td>30</td><td>Luxury</td><td>52</td><td>0.0</td></tr><tr><td>Honda City</td><td>36</td><td>Super Deluxe</td><td>22</td><td>0.0</td></tr><tr><td>Mercedes</td><td>26</td><td>Luxury</td><td>45</td><td>0.0</td></tr></table> Assuming each car runs 1250 kilometers per month with Petrol price 108 rupees per Litre. b) Define a function to find Efficiency (<i>total cost per month</i>) of each car and update into the dictionary. c) Write a code to find the <i>name</i> of the car and its type which have maximum Efficiency.	C-Name	Average (in KM/Litre)	C-Type	C-price (in lakhs)	Efficiency (per month)	Shift Desire	42	Deluxe	13	0.0	Audi	30	Luxury	52	0.0	Honda City	36	Super Deluxe	22	0.0	Mercedes	26	Luxury	45	0.0	[4+ 3 +3]	INT2121.2
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Q 13	Write python code to do following tasks: - Convert the dictionary Car created in Q12 into DataFrame Cars. - Find the Car names whose average is more than 35 KM/Hr and store the result in Series/Dataframe. - Delete the record from Dataframe Cars if the car Type is '<i>Deluxe</i>'. - Delete the field "C-Type" from Dataframe Cars. - Add a new record into Dataframe Cars as ('BMW', 39, 'Luxury', 48, 0.0)	[2+2+2 +2+2]	INT2121.5																									